

it distributes the load uniformly.

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NON - METALLIC MATERIALS . . .

1. Plastics : -

* Definition of Plastics → Plastics is to refer the organic materials of high molecular weight which can be moulded in any desired when subjected to heat and pressure in the presence of a catalyst.

- Resins → Resins are the basic binding materials, which is the major part in compounding of plastic and which undergoes polymerization and condensation reaction during the moulding operation of plastics.

* Formation of Plastics by Addition & condensation Polymerisation by giving -

- ① Addition polymerisation - This reaction forms long chain polymers from small chain monomers without eliminating any atoms or molecules.

Monomers with double or triple bonds are repeatedly added to form polymers. This process is thermodynamically favorable because of its lower activation energy.

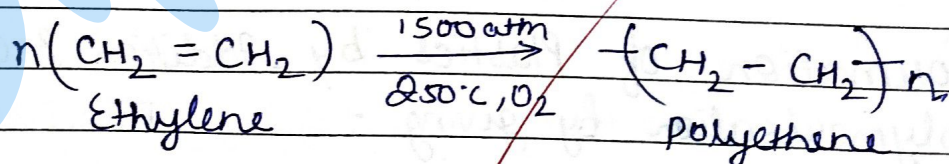
(ii) Condensation polymerization $\rightarrow$ This reaction forms long chain polymers from monomers while eliminating small molecules like water, ammonia, or hydrogen chloride.

- Monomers with two similar or different functional groups undergo repeated condensation reactions to form polymers.

* Preparation and uses of Plastics: -

(1) Polythene $\rightarrow$ It is a rigid, waxy white translucent non-polar material having chemical resistance to strong acid alkalis and salt solution at room temperature.

Preparation: - It is prepared by the addition polymerization of ethylene at 1000-2000 atm. pressure and 250°C , temperature, in the presence of O_2 .



Uses of polythene: -

(1) It is used for making high-frequency insulators

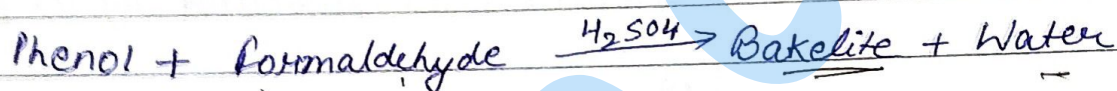
parts, flexible bottles, kitchen and domestic appliance.

(ii) They are used to make toys, sheets for packing materials, tubes-coated wires and cables etc.

(2) Bakelite or Phenol Formaldehyde Plastics :-

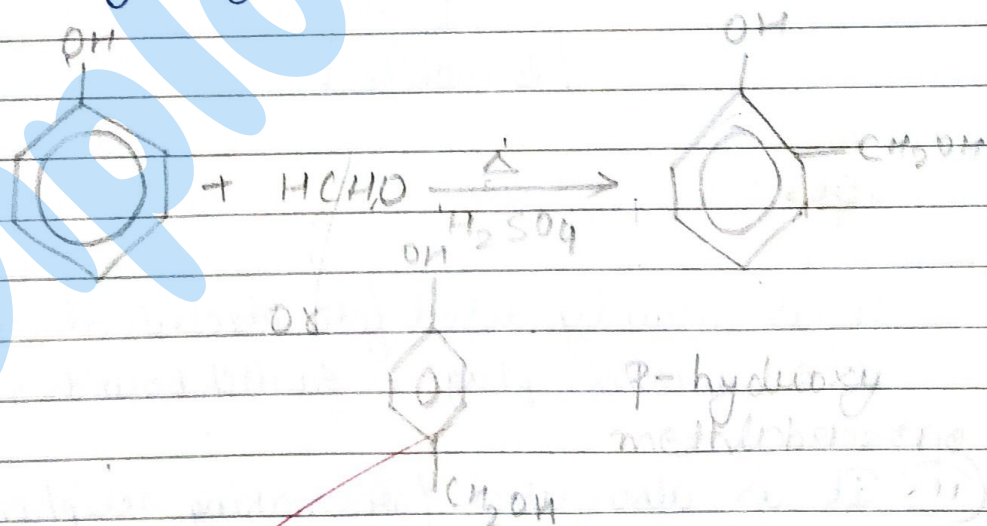
→ It is a rigid, hard substance. It is the first synthetic resin.

Preparation :- It is prepared by the condensation polymerization of phenol and formaldehyde in the presence of acid or alkali.



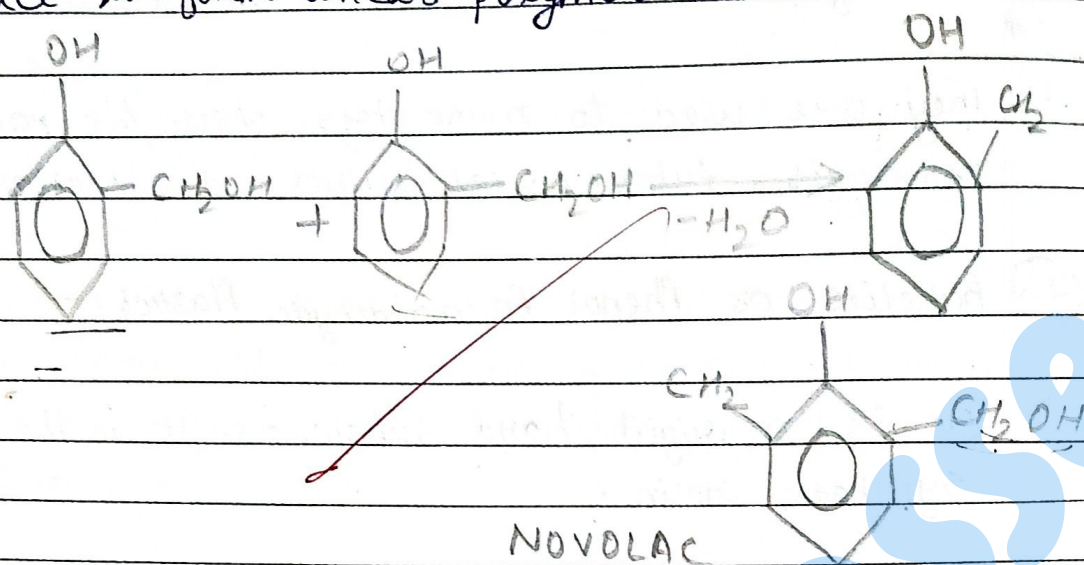
Formation of Bakelite takes place in 3 steps :-

1st step :- Phenol reacts with formaldehyde to produce ortho or para-hydroxy methylbenzene



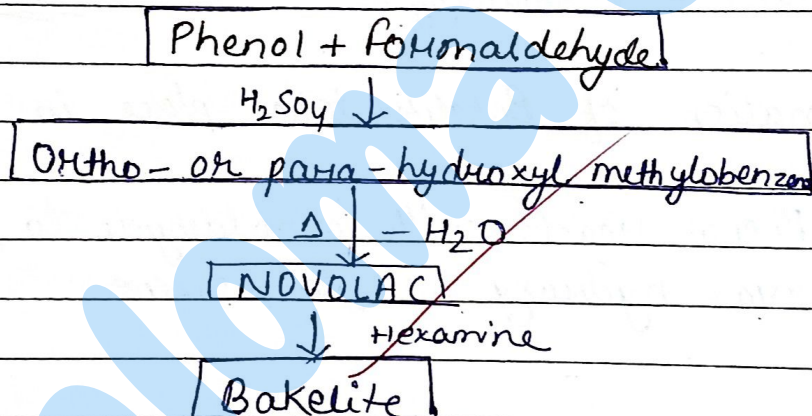
2nd step :- Ortho- and para hydroxy methylbenzene

react to form linear polymer known as NOVOLAC.



Step 3. NOVOLAC on heating with hexamine (hexamethylene tetramine) forms bakelite.

Bakelite preparation is indicated by the following.



Uses : -

- (i) It is mainly used for electrical insulator parts like switches plugs, switch boards, heater handles.
- (ii) It is also used for making telephone parts, cabinets for radio etc and television.

(iii) It is used in ion exchangers, bearings, in paints and in varnishes.

(iv) It is used as binder for grinding wheels.

* Polymerization → Polymerization is a process of uniting or linking together of two or more smaller and simpler molecules of similar or different types with or without the elimination of water etc.

* Types of Plastics —

(1) Thermosoftening → Thermosoftening plastics are those which can be softened on heating and harden on cooling reversibly. These are linear polymers without any cross-linkage in their structure.

→ They possess the ability of deform plastically as the temperature increases due to breaking of the secondary bonds between the individual chains. Example :- nylon.

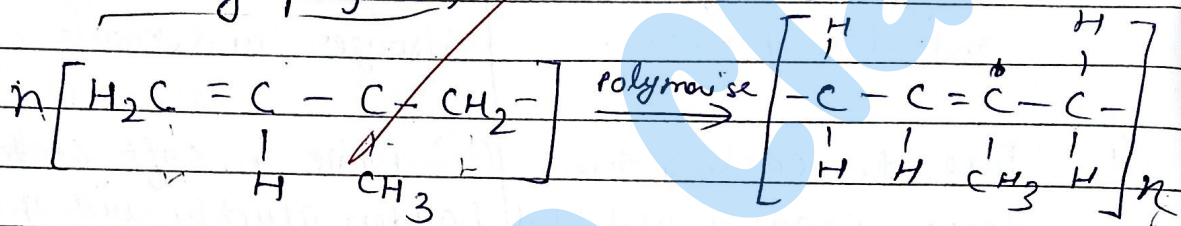
(2) Thermosetting plastics :- Thermosetting plastics are those which during moulding process (by heating etc.) finally get hardened and once they solidify cannot be softened i.e. they are permanent setting type. Example :- polyester, Bakelite etc.

→ As these do not soften on heating hence cannot be reworked. They become harden due to the completion of any left-over polymerization reaction.

These are polymers of smaller molecular weight. Hence PVC is soft whereas Bakelite is thermosetting plastic i.e. the plastic which once hardened cannot be made soft again. It is formed by condensation polymerisation. Hence Bakelite is hard.

Q. RUBBER :-

- Definition of Rubber :- Rubber is a natural polymer of isoprene (polyisoprene), and an elastomer (a stretchy polymer).



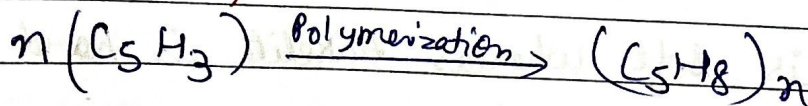
Isoprene (Latex)

Polyisoprene
(Natural Rubber).

- Types of Rubber :-

→ Rubber can be obtained either from the natural source (e.g. latex) or can be prepared artificially.

- i) Natural Rubber :- Latex which is a dispersion of isoprene polymerize to form long coiled chain of polyisoprene i.e. natural rubber.

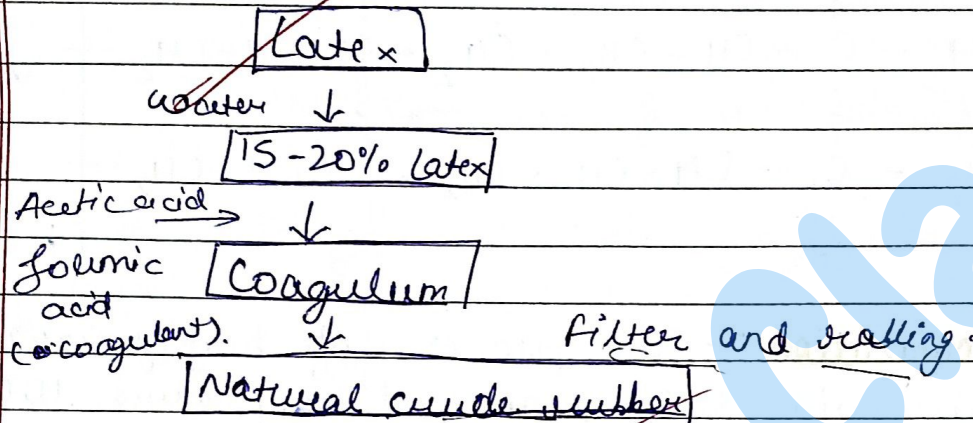


→ Latex is present in the Saps of a wide range of plants such as *Hevea brasiliensis* and *Cynagula* found in tropical and semi-tropical countries such

as India, Indonesia, Ceylon, Malaysia etc.

★ Processing of Natural rubber.

→ Coagulum drained for 2 hrs. further it put into creping machines again convert into sheet.



★ Drawbacks of Natural Rubber :-

→ i) It is plastic in nature.

(ii) It is weak (tensile strength is low 200kg/cm^2).

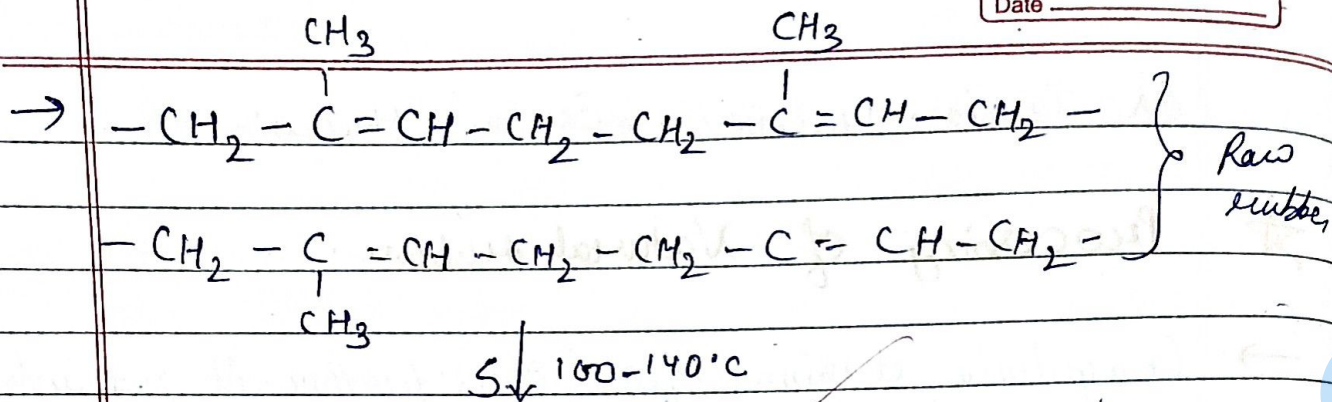
(iii) It has high water-absorption capacity.

(iv) It has little durability.

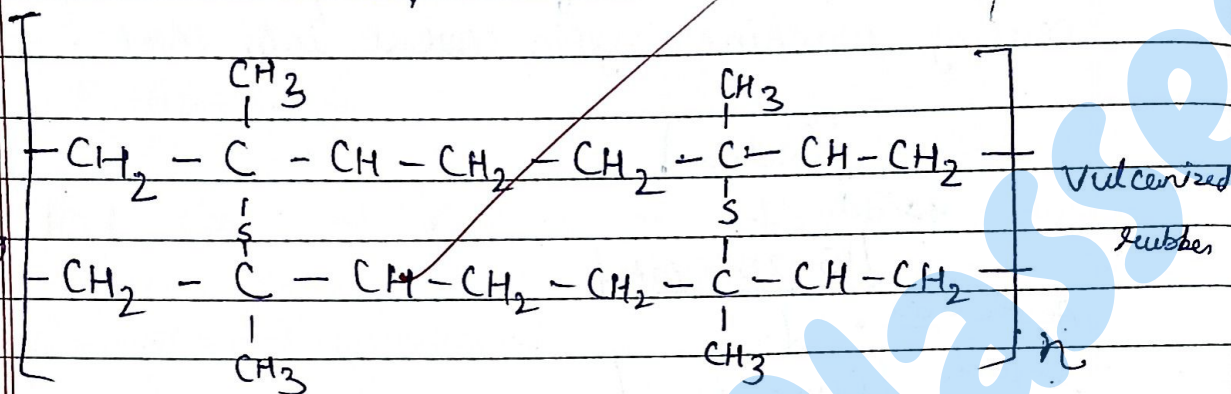
(v) It perishes due to oxidation in air.

(vi) It is non-resistance to non-polar solvent (like vegetable and mineral oils, benzene, etc., -minerals etc).

★ Vulcanization :-



$\xrightarrow[\text{S}]{100-140^\circ\text{C}}$



• Vulcanization :- The process of heating raw rubber with sulphur and at ^{high} temperature $100-140^\circ\text{C}$.

→ To improve the properties of rubber.

e.g. - tyre rubber $\rightarrow \text{S} \rightarrow 3 \text{ to } 5\%$

battery case rubber $\rightarrow \text{S} \rightarrow 30\%$

Properties of vulcanized rubber depends upon:-

- The amount of sulphur used.
- The temperature at which vulcanization is carried out.
- Duration of vulcanization.

→ Rubber containing 32% sulphur is called vulcanite or ebonite or hard rubber, which has no elasticity.

* Characteristics of vulcanized rubber.

→ i) It has good tensile strength and extensibility.

(ii) It has low water absorption capacity.

(iii) It is a better electrical insulator.

(iv) It has lower elasticity.

(v) It is resistant to organic solvents like petrol, benzene, fats and oils etc.

(vi) It has only slight tackiness.

* Synthetic Rubber :-

→ These are the group of high polymers which possess to a more or less than extent the physical properties of natural rubber.

The main synthetic Rubber are :-

(i) Styrene rubber (GR-S) (ii) Nitrile Rubber (GR-A)
or Buna-N.

(iii) Neoprene (GR-M) (iv) Butyl rubber (GR-I)

(v) Thio Kol (GR-P) (vi) Hypalon

(vii) Isocyanate rubber (viii) Silicon rubber.

* Properties of Synthetic Rubber :-

i) Elasticity $\rightarrow$ Elasticity which is opposite of plasticity is measured by determine plasticity of rubber. Plasticity is measured by adding the two (i.e. Softness and ~~un~~ retentivity), the value is high the rubber stock is plastic.

ii) Tack :- Tack is important particularly in blending rubber with other substances to prepare composite rubber articles such as tyres etc. The freshened surface are simply together and then pressed together.

iii) Abrasion Resistance :- Wearing away of the surface by friction is called abrasion. The loss in volume of rubber stock in a definite period denotes the loss on abrasion. The smaller the loss ~~great~~ ~~greater~~ the abrasion resistance of the rubber stock and vice-versa.

* Application of Rubber :-

i) Motor tyres and tubes for automobiles and other vehicles are manufactured from rubber.

ii) Conveyor and V-belts used for conveying many types of goods such as grain, sand, coal etc.

iii) Rubber-lined parts for engineering plants where great resistance to abrasion is required e.g.

cast iron chutes are lined with rubbers.

(IV) Rubber-lined metal tanks and vessels are used in chemical plants to resist the corrosive action of chemicals processed.

(V) Foam rubber is used for bed, pillows, mattresses, cushions etc.

(*) Distinguish b/w Natural & Synthetic rubber.

→ Natural Rubber

Synthetic Rubber

1) It is naturally occurring linear hydrocarbon polymer elastic in nature obtained from latex of rubber trees.

① It is man made an artificial rubber like product obtained by some chemical reactions.

② It is soft and sticky at higher temperature.

② It does not become soft and sticky at higher temperature.

③ It undergoes permanent deformation.

③ It undergoes temporary deformation.

④ It is plastic in nature.

④ It is elastic in rubber.

⑤ It has high tack property.

⑤ It has low tack property.

⑥ It contains isoprene molecule.

⑥ It is obtained from unsaturated substance.

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* (3) Thermal Insulating Materials :-

• Define thermal insulating material.

→ The substances having extremely low thermal conductivity are called thermal insulating material.

• Explain different factors affecting the thermal conductivity of insulators.

→ Thermal insulators depends on two factors:-

(1) Pores

(2) Moistures

(1) Pores :- A large number of fine pores is preferable to having few large ones.

(2) Moisture :- The surface of insulator should be water proof as presence of moisture, increase the thermal conductivity.

• State four characteristics of good insulating material.

→ characteristics of good thermal insulator-

1) Its thermal conductivity should be low.

(2) It should be fire proof.

(3) It should be cheap.

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Material against heat and elasticity -

• Uses :-

- ① In water proof coatings, linoleum etc.
- ② In making base of telephones.
- ③ It is used as lining material for cold storages, refrigerators, bottle - stoppers etc.